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Information Retrieval and Semantic Web. (ENGS4306)

Lab 01 — Boolean Information Retrieval

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Link to github repo

https://github.com/Samarbal/IRSW_Sem1_Lab1

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Lab 01: Boolean Information Retrieval

1. Objective

- The aim of this lab was to build a Binary Incidence Matrix and use AND, OR, and NOT operations to retrieve learning resources. I also explored the difference between matching a query and finding a useful resource.

2. Building the Incidence Matrix

I used 12 learning resources to build the incidence_matrix(). Each row represents a term or metadata tag, and each column represents a resource.

- **1:** The term or tag exists.
- **0:** The term or tag does not exist.

For example, R09 has a value of 1 for topic_python, lang_ar, access_free, and type_notebook. Repeating a word does not change its binary value.

3. Implementing Boolean Operations

I implemented three functions:

- and_vector(): Returns 1 when both values are 1.
- or_vector(): Returns 1 when at least one value is 1.
- not_vector(): Flips 0 and 1.

I also added a check for unequal vector lengths in AND and OR. The provided tests passed (Basic checks passed).

NOT operates within the current collection of 12 resources.

4. Running the Search Queries

Query	Predicted IDs	Actual IDs
Q1	R01, R09	R01, R09
Q2	R04, R05	R04, R05
Q3	R03, R06, R09, R11	R03, R06, R09, R11
Q4	R08	R08

Understanding the Results

- **Q1:** R01 and R09 are free Arabic beginner Python resources.
- **Q2:** R04 and R05 are free, non-advanced Information Retrieval resources.
- **Q3:** Four free Python or NLP notebooks matched.
- **Q4:** Only R08 matched the free web-resource conditions.

For Q2, R04 was included because it satisfied all conditions, while R09 was excluded because its topic is Python rather than Information Retrieval. With AND, failing one condition is enough to exclude a resource.

5. Comparing Search Results

Search	**Results**
python	R02, R10
بايثون	R01, R03, R09, R11, R12
Both words (OR)	R01, R02, R03, R09, R10, R11, R12
topic_python	R01, R02, R03, R09, R10, R11

Searching in one language misses resources using the other language. R12 also matched بايثون, although it is not a practical Python learning resource.

The manually assigned topic_python tag connects relevant Arabic and English resources without automatic translation. This shows that a word match does not always mean a useful result.

6. Choosing a Useful Resource

For a beginner who understands Arabic and wants to practice Python within one hour, I would choose **R09 instead of R01**.

R09 is a notebook with practical exercises and takes 60 minutes, while R01 is a video tutorial.

7. My Own Search Query

Information need: Free English resources about Information Retrieval that are not advanced.

Query: topic_ir AND lang_en AND access_free AND NOT level_advanced

Prediction: R04

Actual result: R04

R04 matched all conditions, while R06 was excluded because it is an advanced resource.

8. Additional Tests

Test 1 — Repeated Terms: I repeated "Python" in a resource's title and summary. The incidence value remained 1 because the matrix records presence, not frequency.

Test 2 — NOT on an All-Zero Vector: Applying NOT to 12 zeros returned 12 ones, meaning all resources were included in the result.

9. Conclusion

This lab showed how Boolean Retrieval uses binary values to find matching resources. However, it does not rank results by relevance, understand query meaning, or automatically handle Arabic–English vocabulary differences.